

JURAJ BEZÁK

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FINANCIAL DATA SOLUTIONS ARCHITECT | RISK MODELING | ANALYTICS SPECIALIST

Financial Data Solutions Architect with 10+ years of enterprise experience in risk modeling platforms and regulatory reporting within the insurance sector. Specialized in designing scalable, R-driven data pipelines and auditable workflow execution engines for Solvency II and management reporting. Bridging actuaries, IT, and regulatory stakeholders to deliver high-integrity production architectures.

SKILLS

Financial Data Pipelines | Solution Architecture | Stakeholder Management

Risk Modeling | Regulatory Reporting | Solvency II Regulation | R (Advanced / Package Ecosystems) | SQL

KEY PROJECT HIGHLIGHTS

- **Engineered end-to-end risk and financial data flows** across multi-entity production cycles using an enterprise workflow execution engine. Ensured full auditability, reproducibility, and four-eyes principle governance across 12 legal entities and 30 branches spanning 7 national regulators and 17 countries.
- **Rapidly resolved a mission-critical platform failure on regulatory submission day** under strict deadline pressure, averting severe regulatory penalties; awarded the **Zurich Silver Award** and subsequently appointed Platform Owner.
- **Designed a taxonomy-driven data model with an automated pre-reporting validation framework** (covering DPM taxonomy and custom business rules), significantly reducing manual verification cycles and operational support overhead.

EXPERIENCE

ZURICH INSURANCE COMPANY LTD, Risk Modeling Platform, Bratislava, Slovakia / *October 2018 – June 2026*

Senior Solution Design Specialist, June 2024 – June 2026

Provided business and technical guidance in cross-application risk models, designed enterprise financial data flows, and mentored analytics developers. Accountable for aligning complex business and regulatory requirements with scalable data architecture.

- Led the cross-country migration of a large branch, ensuring business continuity and operational stability.
- Assumed Release Manager responsibilities; eliminated manual deployment overhead by automating audit-ready CI/CD pipelines in R integrated with Jira and Confluence APIs.
- Architected and implemented data pipelines using an in-house Workflow Execution Engine (enterprise data platform comparable to Palantir Foundry).

Senior Application Developer, May 2020 – May 2024

Led technical design, optimization, and automated release management of insurance risk models (Life and Non-Life) and data workflows for management and regulatory reporting (QRT, NST, FSR, RSR, ORSA, SFCR).

- Earned a **Zurich Silver Award** for resolving a submission-day platform crash and assuming platform ownership (internal platform with Regnology Insurance as final reporting tool).
- Restored a legacy reporting application 3 weeks before deadline, avoiding a delayed replacement project.
- Built a data interface framework mapping technical objects onto the EIOPA Data Point Model (DPM).

Application Developer, October 2018 – May 2020

Focused on development and maintenance of financial risk data flows, aggregation pipelines, and statistical modeling tools.

- Built financial risk models using R and SQL, automating regulatory reporting in XML and XBRL formats.
- Led legacy stack modernization: migrated codebase from S-PLUS (S++) to R, version control from SVN to Git, and CI/CD automation from Bamboo to Jenkins.
- Mentored junior developers in R programming, data workflows, and delivery best practices.

ACCENTURE, Bratislava, Slovakia / April 2016 – October 2018

Senior Application Developer

- Consulted on and developed financial risk models in S-PLUS (S++)/R.
- Delivered regulatory risk reporting solutions integrated with the Regnology Abacus Insurance platform.
- Transitioned application ownership from external contractors to an internal team, introducing auditable documentation, industry-standard processes, and a new remote working model.
- Established a structured onboarding program and mentored new team members.

SLOVAK UNIVERSITY OF TECHNOLOGY IN BRATISLAVA, Department of Land and Water Resources Management, Bratislava, Slovakia / September 2010 – July 2015

PhD Researcher (Landscape Engineering), 2010 – 2015

Conducted advanced research in hybrid data-driven modeling and machine learning for high-dimensional hydrological forecasting.

- Developed ensemble machine learning models (Gradient Boosting, Random Forest, SVM, MARS, Neural Networks, Generalized Additive Models) for high-dimensional hydrological modeling and flood forecasting.

ADDITIONAL RELEVANT EXPERIENCE

INFOSTAT - Institute of Informatics and Statistics, Bratislava, Slovakia / September 2015 – December 2015 **Macroeconomic Analyst**

EDUCATION

PhD in Landscape Engineering

Slovak University of Technology in Bratislava, Faculty of Civil Engineering, Bratislava, Slovakia

Master's Degree (Ing.) in Landscape Engineering

Slovak University of Technology in Bratislava, Faculty of Civil Engineering, Bratislava, Slovakia

PROFESSIONAL DEVELOPMENT & CERTIFICATIONS

- **Zurich Silver Award** – Recognition for critical platform recovery & process improvement
- **Winning in the Market** – Professional training program on strategic relationship and stakeholder management (6 months)
- **Riva Academy** – Iris Insurance Simulation: Strategic business simulation for the insurance sector

COMMUNITY INVOLVEMENT

- **External PhD Thesis Committee Examiner / Defense Reviewer** — “Contribution to the Water Management Solution for Irrigation Reservoirs”; evaluated machine learning and data-driven methodologies in hydrological modeling.
- **Academic Dissemination & Research Publications** in data-driven hydrological modeling and machine learning.
- **Technical Advisor** — Provided consulting on IT and data architecture solutions for the Faculty of Civil Engineering, STU in Bratislava.

ADDITIONAL INFORMATION

- **Languages:** English (Professional), Slovak (Native), German (Basic / A2)
- **Cross-functional Leadership:** Expert at bridging technical developers, actuaries, and executive stakeholders
- **Core Technical Stack:** R (Advanced / Package Ecosystems), SQL, Atlassian Suite (Jira, Confluence, Bitbucket, Bamboo), Regnology Insurance (Abacus), FS Assist, Solvency II regulation/reporting (DPM / DTM)
- **Supporting Tools & Technologies:** Python (pandas, scikit-learn), Power BI, Tableau, CI/CD Automation (Jenkins & Groovy), Docker, Kubernetes, Artifactory, AI-assisted engineering (GitHub Copilot, Claude, Antigravity, Codex), VS Code, RStudio, Positron